DRILL BITS

THE COMPLETE REFERENCE — anatomy, types, materials, sizes, speeds and shop craft for cutting round holes in anything

0.1–80+ mm	diameter range — from hair-thin to bottle-bore
100,000 rpm	micro drills on air-bearing spindles
90–135°	point angle — flat wood to tough hardened steel
5 materials	HSS · cobalt · carbide + TiN / TiAlN coatings
2/3/4 flutes	helical grooves — chip evacuation paths

1 · THE TOOL

ANATOMY OF A TWIST DRILL

Every drill cuts with lips and guides chips up helical flutes

Shank — the end held in the chuck; straight (round, for a keyed/keyless chuck) or **taper (Morse)** for large drills and spindles. Some have flats or a tang to stop slipping.

Body — the long ground cylinder. Runs slightly relieved toward the shank so only a thin **margin** (a land strip) touches the hole wall — this keeps friction low and the hole true.

Flutes — spiral (helical) grooves that do double duty: they form the cutting edges and are the **chip-removal channels**. Standard **right-hand helix** lifts swarf out as the drill rotates clockwise. **Spiral angle** (helix): ~25–30° general, low (10–15°) for brass/copper to stop bit grabbing, high (~35–40°) for deep soft-metal and wood holes that flood with chips.

Lips (cutting edges) — the two (or three/four) sharp edges ground on the point that actually cut. Each lip cuts like a tiny scraper shearing material into a chip.

Point — the cone-shaped cutting end. Made of the **web** (the central core of metal left between the flutes, thickest near the point for strength), the **chisel edge** (the flat central ridge that does no true cutting but pushes material — the reason drills wander), and the **margins** (the narrow lands that guide and centre the bit). Thinning the web upsets the chisel and improves self-centring.

Tip (point) angle — the included angle of the cone: **118°** for general steel/wood, **135°** for stainless and hardened steels, **90°** for brass/plastic, ~60° for very soft — a flatter point resists drill 'pulling in'.

4 · METALLURGY

DRILL MATERIALS

Choose the body AND the coating by the workpiece

Material	Strength & the 'when'
HSS	High-Speed Steel: tough, heat-resistant to ~600°C, dulls gracefully, cheap, easy to sharpen. The everyday choice for wood, mild steel, aluminium, plastic.
HSS-Co (Cobalt)	M2/M35/M42 with 5–8% cobalt — keeps its edge at higher temperature. For stainless steel, titanium, and high-heat production drilling.
Carbide solid	Sintered tungsten carbide — ~3× harder than HSS, cuts 3–5× faster, but brittle. For production, aluminium/carbon fibre, and CNC. Must be rigid and well-cooled
Carbide-tipped	A tough steel body with carbide cutting edges brazed on. Cheaper than solid, less brittle — masonry bits, annular cutters.
TiN / TiAlN / TiCN coated	Hard ceramic films (gold/silver/blue): lower friction, higher wear and heat resistance. TiAlN shines on dry machining of steel; TiN general-purpose; TiCN great or
PKD / PCB / diamond	Polycrystalline-diamond tipped or diamond-core — the hardest, for tiles, glass, stone, composites and carbide workpieces.

Rule of thumb: soft workpiece + rough job → HSS; hard/tough workpiece or high speed → cobalt or carbide; abrasive/brittle (tile, glass) → diamond-tipped.

5 · THE POINT

DRILL POINT GEOMETRIES

The shape of the point sets the hole's accuracy

Standard 118° — the all-round point cone. Best compromise between cutting speed and lifespan for general steel and wood; needs a centre punch so it does not wander.

Split point (self-centring) — a nick ground across the web cuts the chisel edge away, creating two extra lips. The bit starts on its own — no punch, no walking. Essential for handheld and CNC; the modern default for twist drills.

135° point — steeper, shorter cone. Withstands the high pressure and heat of stainless and hardened steels; less likely to 'pull in' or grab.

Rake (helix / spiral angle) — the angle of the cutting face. Higher rake cuts freer (soft materials, wood); low rake resists grabbing in brass/copper; negative rake is for carbide on hard stuff.

Parabolic flute (deep-hole) — extra-wide, rounded flutes that cannot clog; made for deep holes (4–10× diameter) where chips must flow freely without packing.

Brad point — a centre spur + two rim spurs; the spur centres the bit and the spurs score a clean rim, so exit holes stay un-torn. For wood and dowel work.

Dowel bit — brad-point or 90°-point with centre spur, made to bore clean straight dowel holes in end grain.

3 · THE FAMILY

DRILL TYPES — WHICH BIT FOR WHICH HOLE

13 workhorses — match the bit to the material, the depth and the hole quality you need.

Drill type	What it is	Best use
Twist / jobber	All-round HSS spiral drill; the 'normal' drill, sets 1–13 mm.	Wood, metal, plastic — the default bit.
Stub	Short, thick twist with a rigid body — less flex and wander.	Short holes needing accuracy and stiffness.
Long / silver & Deming	A 1/2" shank twist drill with a reduced cutting diameter and extra-long flutes.	Reaching deep/awkward spots; 'aircraft' patterns for long reach.
Centre drill	Short stubby bit that drills a combined pilot + 60° countersink in one pass.	Starting lathe centres and accurate pilot holes.
Step drill (unibit)	One cone of stepped diameters; edges shear sheet cleanly, no grabbing.	Thin sheet metal, plastic, enclosures — one bit, many sizes.
Countersink	Cone-shaped cutter matching screw-head angle (82° standard, 90° for metric).	Flush screw heads in wood and metal.
Counterbore	Flat-bottomed recess for the head of a socket (Allen) cap screw.	Making a flat seat for a bolt head.
Spade (paddle)	Flat paddle with a centre point; bores wide holes fast, rough exit.	Quick, rough wood holes; back with scrap.
Masonry	Carbide-tipped, blunt flat head that smashes aggregate.	Brick/block/concrete — ONLY with hammer action.
Gun / straight-flute	A single straight-flute bit with one cutting edge; chips eject straight out.	Deep, clean holes in soft metals and plastics.
Indexable insert	Carbide insert (like a lathe tool) bolted on a steel body.	Large holes in production machining; replaceable edge.
Core / annular cutter	Hollow ring with teeth that cuts an annular slug out of solid steel.	Large holes in plate/beams with a magnetic (mag) drill.

6 · SIZING

DRILL SIZES & NUMBERING

Four naming systems — always read the shank engraving

1. Fractional (inch) — diameters in fractions of an inch (1/16" up to 1"+), by 1/64" steps. Shops and USA-standard.

2. Metric — diameter in millimetres (0.1 to 80+ mm), often 0.5 or 0.1 mm steps; the world standard, code-numbers from the ISO series (e.g. HSS M2, Ø 8.5).

3. Wire gauge (Number) 1–80 — the older numbered system; the LARGER the number, the SMALLER the diameter. #80 ≈ 0.0135" (hair-thin), #1 ≈ 0.228". Originated from Stubs wire gauges — used for tiny model and PCB drills.

4. Letter (A–Z) — 0.234" (A) up to 0.413" (Z), filling the gap between the numbered (small) and fractional (large) sizes. A 'Q' drill ≈ 0.332", etc.

Diameter ↔ size chart — a full drill-bit chart lists, per size: exact diameter, area, tapping-drill and clearance-drill mate for each screw thread, and metric/inch cross-reference. Printed wall charts are the workshop standard.

Collet code — **DIN 338** = jobber (normal length), **DIN 340** = long, **DIN 1897** = stub. Shank diameter = drill diameter for most (unless reduced).

7 · SPEED & FEED

CUTTING SPEED & SPINDLE RPM

The two numbers that decide whether a bit cuts or burns

Cutting speed V (m/min) — how fast the cutting edge travels. Too slow → it rubs and work-hardens; too fast → the edge burns out. Pick V by material, then convert to spindle RPM.

Spindle speed formula: n = (1000 × V) ÷ (π × D) — n in rpm, V = cutting speed (m/min), D = drill diameter (mm). Smaller drill → much faster spindle.

Workpiece	V m/min (HSS)
Aluminium / brass	60–100
Mild steel	25–35
Stainless steel	8–15
Cast iron	18–25
Copper	30–50
Wood / plastic	70–120

RPM by diameter (mild steel, V=30): Ø1 mm → 9,500 rpm · Ø3 mm → 3,200 · Ø6 mm → 1,600 · Ø10 mm → 960 · Ø20 mm → 480 · Ø80 mm → 120 rpm.

Feed per revolution — how far the bit advances each turn; bigger drills take more (Ø6 ~0.08 mm/rev) and smaller take less. Too much feed → chipped lips; too little → rubbing. **Peck drilling** = retract regularly to clear chips and cool the tip.

8 · COOLING

COOLANT & CHIP REMOVAL

Friction creates heat; chips must escape or the hole goes wrong

- **Flood coolant** — a heavy stream washes over the bit; best cooling + chip clearing. For production metals.
- **Mist (air-oil spray)** — light atomised coolant; used on manual/low-volume jobs, good for aluminium.
- **Through-tool coolant** — coolant pumped down holes in the drill body to the cutting edge; for deep, hard, production holes.
- **Paste/oil** — cutting oil or wax on the point for hand-drilling steel and stainless.
- **Dry / no coolant** — cast iron and wood run dry; brass and plastics often run dry or lubricated lightly.

Chip packing — when long strings of swarf clog the flutes, friction and heat spike and the drill can jam or break. The fix is **pecking**: drill a little, retract, let the flutes re-grab and pull chips out, repeat.

Deep holes (>3× diameter) — always peck; use parabolic-flute or gun drills; keep coolant on the point; watch for chip 'birds-nesting' around the bit.

9 · TROUBLESHOOTING

DRILLING DEFECTS & FIXES

Every failure has a cause — read the symptom, fix the physics.

Defect	What you see	Main cause	Fix
Wandering / off-centre	Hole starts at an angle or tears sideways	Chisel edge pushes; no centre punch; 118° point; flexing long bit	Centre punch; split-point bit; pilot hole; stiffer stub bit
Oversize hole	Hole measures larger than the drill Ø	Unequal lip grind; worn margins; off-centre point; wobbling spindle	Sharpen/check lips equal; new bit; true the spindle
Chatter	Squeal, rough wall, oval hole	Workpiece not clamped; too much feed; long flexing bit; wrong speed	Clamp work; steady feed; stub bit; correct rpm
Work hardening	Surface gets harder, bit won't cut	Drill rubbed (too slow) instead of cutting stainless/aluminium	Keep feed aggressive; sharp bit; correct high speed
Burned (blue) edges	Lip goes blue/soft, won't hold edge	Too fast / no coolant / dull bit	Slow the spindle; add coolant; sharpen or replace
Broken bit	Drill snaps in the hole	Excess feed; chip jam; deep hole no peck; hole not supported	Peck; relieve pressure; back plate; coolant; pilot hole
Burr (ragged) exit	Metal 'tears' out the far side	No backing; bit bursts through the last skin	Backing plate / clamp scrap under work; drill then deburr

10 · SHOP CRAFT

DRILLING OPERATION ADVICE

Preparation and technique beat brute force

- **Centre punch** first — a dimple gives the bit a starting seat so it cannot wander on the smooth surface.
- **Pilot hole** — drill a small hole first, then open it out; guides the big bit, cuts force, eases heat.
- **Clearance vs tapping hole** — a clearance (through) hole is the OVERSIZE hole a screw passes through freely; a tapping hole is the UNDERSIZE hole that threads are cut in (e.g. for M6: clearance ~6.2 mm, tap ~5 mm).
- **Countersink depth** — stop when the screw head is just flush or ~0.5 mm below; drill countersink + pilot in one bit or two operations.
- **Backing plate** — clamp a scrap under the work so the bit exits into scrap — clean holes, no breakout tear-out, protects the bench.
- **Start slow, then steady** — begin at low speed with light pressure for a true start, then feed firmly and continuously.

Sharp bit + correct speed + steady feed + coolant = clean hole every time.

11 · COMPARE

DRILL vs REAMER vs TAP

Three tools, three different jobs on a hole

Tool	What it does	Hole
Drill	Cuts the hole from solid — rough, slightly oversize (by design).	Rough
Reamer	Slicks an EXISTING drilled hole to an exact size and finish (adds ~0.1–0.3 mm).	Precise
Tap	Cuts internal THREADS into a pre-drilled tapping hole.	Threaded

HOLE-MAKERS — **twist drill**: small, precise, deep · **step bit**: many sizes in one, thin sheet · **hole saw**: big circles in wood/metal quickly · **annular cutter**: large clean holes in thick steel plate (mag drill).

Pick: **<13 mm & precise** → twist · **thin sheet, many sizes** → step · **big soft holes** → hole saw · **big holes in plate** → annular/core cutter.

12 · STAY SAFE

HANDLING DRILLS SAFELY

The rules that keep fingers off the bit

- **Eye protection** — swarf, dust and sparks fly at speed; always wear goggles or a face shield.
- **No gloves near the drill** — a glove can be caught and wrapped around the spindle, dragging your hand in. No loose sleeves, ties or long hair.
- **Secure the workpiece** — clamp or vice it firmly; a grabbing bit can spin the piece into your face.
- **Remove chips with a brush** — never flick swarf with bare fingers; it is razor-sharp and hot.
- **Never force a dull bit** — pressure on a dull edge builds heat and snaps the bit; sharpen or replace.
- **Unplug / lock before changing** — always remove the drill or swap bits with the power isolated.
- **Let the bit stop** — never grab a spinning bit, chuck or work to slow it.

First rule: sharp, correct-speed, clamped, and you are clear of the cut.

13 · WORDS

GLOSSARY — 14 KEY TERMS

Speak the language of the drill

Shank — the end gripped by the chuck (straight or Morse taper)

Lip — the sharp cutting edge at the point that shears the material

Web — central metal core between the flutes; thickest at the point

Margin — narrow land on the body that guides and centres the bit

Cutting speed V — edge speed in m/min — pick by material, then set rpm

Pilot hole — small hole drilled first to guide a larger bit

Reamer — tool that sizes/smooths an existing drilled hole

Flute — spiral groove that forms the cutting edge and ejects chips

Point — cone-shaped cutting end; its angle sets how it bites

Chisel edge — flat central ridge that pushes material and causes wander

Helix — spiral angle of the flute; sets chip flow and rake

Peck drilling — retracting regularly to clear chips and cool the tip

Countersink — cone recess so a screw head sits flush

Tap — tool that cuts internal threads into a tapping hole